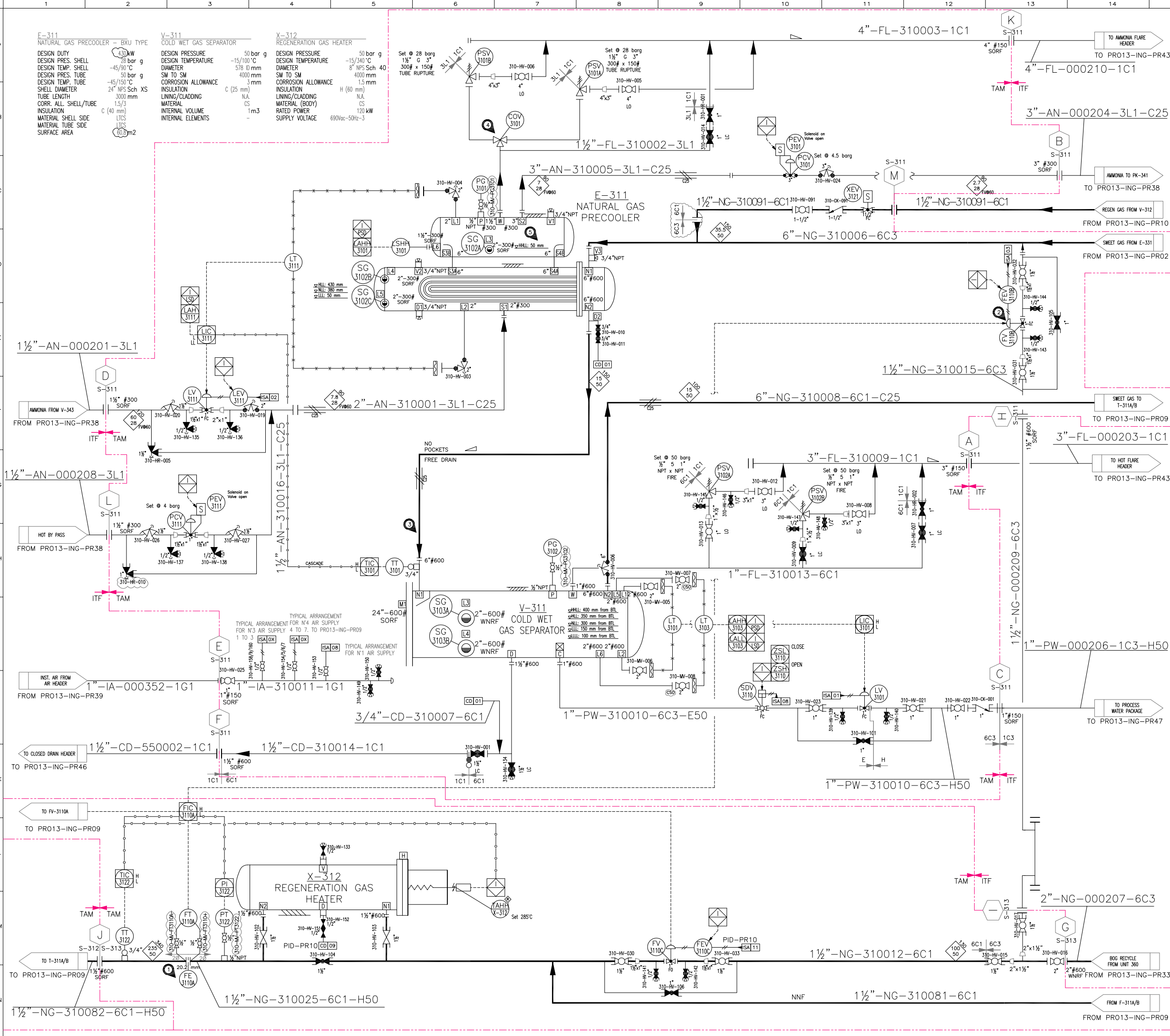


E-311 NATURAL GAS PRECOOLER - BXU TYPE	V-311 COLD WET GAS SEPARATOR	X-312 REGENERATION GAS HEATER
DESIGN DUTY 430 kW	DESIGN PRESSURE 50 bar g	DESIGN PRESSURE 50 bar g
DESIGN TEMP. SHELL -45/90 °C	DESIGN TEMPERATURE -15/100 °C	DESIGN TEMPERATURE -15/340 °C
DESIGN TEMP. TUBE -45/150 °C	DIAMETER 578 ID mm	DIAMETER 8" NPS Sch 40
SHELL DIAMETER 3000 mm	SM TO SM 4000 mm	SM TO SM 4000 mm
TUBE LENGTH 15/3	CORROSION ALLOWANCE 3 mm	CORROSION ALLOWANCE 1.5 mm
INSULATION LTC5	LINING/CLADDING CS	INSULATION H (60 mm)
MATERIAL SHELL SIDE LTC5	MATERIAL VOLUME 1m3	LINING/CLADDING N.A.
MATERIAL TUBE SIDE LTC5	INTERNAL ELEMENTS -	MATERIAL (BODY) CS
SURFACE AREA 60.9 m2		RATED POWER 120 kW
		SUPPLY VOLTAGE 690Vac-50Hz-3



15	16	17	18		
REFERENCE DOCUMENTS					
P&ID LEGEND		PRO13-ING-PRO0 21044-PR-PID-00000			
PROCESS FLOW DIAGRAM		PRO10-ING-PRO1/02/03/04 21044-PR-PFD-00001			
UTILITY FLOW DIAGRAM		PRO10-ING-PR11/12 21044-PR-UFD-00001			
MATERIAL SELECTION DIAGRAM		PRO08-ING-FR01 21044-PR-MSD-00001			
ISA 0X INSTRUMENT SUPPLY AIR CD 0X CLOSED DRAIN TO DRAIN HEADER NNF NORMALLY NO FLOW					
NOTES					
1CONDITIONING 4 HOLES ORIFICE PLATE. 2FIRESAFE GLOBE VALVE. 3THERE IS NO REAL LINE BETWEEN E-311 AND V-311, NOZZLES N1 ON V-311 AND N2 ON E-311 WILL BE CONNECTED FLANGE TO FLANGE. 4CHANGE OVER VALVE WITH LOCK DEVICE. 5HHLL TO BE MEASURED FROM E-311 SCRUBBER BTL. 6DESIGN AND EXECUTION OF ELECTRICAL TRACING BY ITF.					
----- TAM SUPPLY LIMIT					
6	30 Mar 2026	APPROVED FOR CONSTRUCTION	TAM (JB)	TAM (MM)	LOAMPLONE
5	01 Aug 2025	APPROVED FOR CONSTRUCTION	TAM (LD)	TAM (MM)	LOAMPLONE
4	24 Jun 2024	APPROVED FOR CONSTRUCTION	TAM (LD)	LOAMPLONE	WBOULLAU
3	18 Aug 2023	ISSUED FOR APPROVAL	TAM (LD)	LOAMPLONE	WBOULLAU
2	08 Nov 2022	ISSUED FOR APPROVAL	TAM (LD)	LOAMPLONE	WBOULLAU
1	21 Jul 2022	ISSUED FOR APPROVAL	TAM (ES)	LOAMPLONE	WBOULLAU
0	28 Jun 2022	ISSUED FOR APPROVAL	TAM (ES)	LOAMPLONE	WBOULLAU
Rev.	Date	Description	Draft	Check'd	Approved
MANAENERGY		Customer: MANAENERGY	OFFICES: Mahaq Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco		
ITALFLUID		Contractor: ITALFLUID	OFFICES: VIA VIVALDI,4 MONTE SILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742		
ITALFLUID		Engineering: ITALFLUID	OFFICES: VIA VIVALDI,4 MONTE SILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742		
EPS		Technology Vendor: TECHNICAL AMERICA INC	OFFICES: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540		
TITLE: LNG TENDRARA PLANT			OBJECT: P&ID DEHYDRATION UNIT 310 SKID S-311/S-313		
TYPE	SCALE	DOCUMENTS ID:	VENDOR DOC. CODE		SHEET/OF
A1	N/A	PRO13-ING-PRO8	21044-PR-PID-31001		1/1